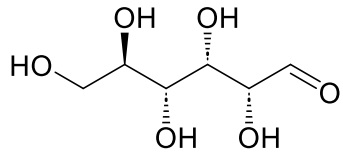
Answer **all** the questions in the spaces provided.

1 Investigation of redox reactions involving D-glucose

D-Glucose is a sugar with molecular formula $C_6H_{12}O_6$, which can react directly or indirectly with potassium manganate(VII), $KMnO_4$.



D-glucose

(a) **FA 1** is a solution of potassium manganate(VII), $KMnO_4$.
FA 2 is a 1.0 mol dm^{-3} sulfuric acid solution, H_2SO_4.

You are also provided with a solution of D-glucose and bench reagents.

Perform the tests described in Table 1.1 and record your observations in the table.

Table 1.1

test		observations
(i)	Add 1 cm depth of **FA 1** to a test-tube. Add an equal volume of aqueous sodium hydroxide. Add 1 drop of glucose solution and shake thoroughly. Allow to stand for 10 minutes, with occasional shaking. Continue with the remaining parts of Question 1.	
(ii)	Add 1 cm depth of **FA 1** to a boiling tube. Add 2 cm depth of **FA 2**. Add 1 cm depth of glucose solution and warm gently. Shake well to mix.	

[2]

(b) Manganese species exist in ions of different oxidation states and colours.

Suggest the identity of the **final** manganese-containing species in **(a)(i)** and in **(a)(ii)**.

final manganese-containing species in **(a)(i)**.....

final manganese-containing species in **(a)(ii)**.....

[1]

(c) In **(a)(i)**, D-glucose, $C_6H_{12}O_6$, reacts with $KMnO_4$ in alkaline medium to give D-gluconate, $C_6H_{11}O_7^-$.

(i) Name the type of reaction that D-glucose undergoes in **(a).**

..... [1]

(ii) Draw the structure of the D-gluconate ion.

[1]

[Total: 5]